Fiscal Year:	FY 2023	**Task Last Updated:**	FY 09/15/2022
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Effects of Low-Dose Radiation on Neurovascular Remodeling and Blood-Retina Barrier Function in Mice, Rats and Rabbits		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (2) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-Crew Health (FLAGSHIP, OMNIBUS). Appendix A-Omnibus, Appendix B-Flagship
Start Date:	12/14/2017	**End Date:**	09/30/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2023 per C. Ribeiro/JSC and NSSC information (Ed., 4/7/22) NOTE: End date changed to 09/30/2022 per NSSC information (Ed., 8/13/21) NOTE: End date changed to 09/30/2021 per NSSC information (Ed., 5/20/21) NOTE: End date changed to 02/28/2021 per NSSC information (Ed., 9/21/20) NOTE: End date changed to 11/30/2020 per NSSC information (Ed., 8/8/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sun, Shu-Wei Ph.D. (Loma Linda University)		
Grant/Contract No.:	80NSSC18K0310		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	A recent report shows that more than 50% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. To date, the mechanisms behind these effects are not fully understood. These visual disturbances have been hypothesized to be related to increases in intraocular (IOP) or intracranial pressure (ICP). Modeling study showed that compromise in the integrity of the vascular blood-brain barrier (BBB) would serve to elevate ICP. Damage to microvascular endothelial cells induced by oxidative stress is one of the major factors thought to underlie the increased permeability of the blood-retinal barrier (BRB), which maintains a similar function in the eye for regulating intraocular pressure as the BBB in the cranium. Our knowledge about low-dose radiation risks with long-term spaceflight on retinal vasculature and their potential role in altering the BRB is very limited and inconsistent. The objectives of this application are to characterize the impact of radiation doses and qualities on retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and alteration of cell-cell interactions in the function of blood-retina-barrier (BRB). Archived ocular tissues samples at dose ranges from 0.1 Gy to 1 Gy with multiple particle types and energies across multiple animal species will be available for sharing from three NASA funded investigators. We will evaluate radiation–induced retinal vascular and tissue remodeling at multiple time points. Our study will elucidate the spectrum of radiation-induced structural and functional alteration in retinal BRB function and the mechanisms by which these changes are mediated. Together, our unique, integrative, and quantitative activities with advanced MRI imaging techniques will provide insight into the molecular mechanisms and pathways of low-dose radiation-induced mitochondrial oxidative damage on retinal cell-cell communication, BRB integrity, and vascular remodeling. The data from established model will provide unique criteria for risks of functional detriments associate with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The data from established model will provide unique criteria for risks of functional detriments associated with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures.
Task Progress:	We received archived ocular tissues in mid-2021 from six-month-old male and female CD1 mice that were either sham irradiated or received whole-body 5-ion simulated galactic cosmic radiation (GCR) exposure at a dose of 0.5 Gy. At four months following irradiation, mice were sacrificed and ocular tissues were collected for histochemical analysis. Fixed ocular samples were collected, sectioning for histochemical analysis for identifying biomarkers for oxidative stress, apoptosis, and blood-retina-barrier (BRB) function; BRB integrity was evaluated with biomarkers aquaporin-4 (AQP-4), a water channel protein, tight junction (TJ) Zonula occludens-1 (ZO-1), and adhesive molecule, platelet endothelial cell adhesion molecule-1 (PECAM-1). A significantly increased expression of AQP-4 was observed in the retina following GCR exposure compared to controls ($p<0.05$) with more pronounced changes observed in the female mice over males. There was also a significant increase in the expression of PECAM-1 and a decrease in the expression of ZO-1 in the retina of irradiated groups compared to controls. Our study revealed that exposure to low-dose ionizing radiation induced cellular oxidative damage that may alter retina structure and BRB integrity. The results also reveal some sex-related differences in the radiation response. Our data was accepted for presentation at the 2022 meeting of the American Society for Gravitational and Space Research (ASGSR 2022). Our data was also presented at the 44th Scientific Assembly of the Committee on Space Research (COSPAR) in July 2022 – Athens, Greece. One paper titled, "Low dose space radiation-induced effects on the mouse retina and blood-retinal barrier integrity", is published in Acta Astronautica 2022 October; 199:412-419. [Ed. Note: see Bibliography]. Augmentation funding has been awarded to participate in a spaceflight study (MHU-8) to evaluate mouse retina and optic nerve head changes following spaceflight. The flight study is currently scheduled in February 2023. As a funded NASA Human Research Program (HRP) collaborator, preparation and in-depth discussion for the study and dissection details, especially related to the ocular tissues, are ongoing.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Abstracts for Journals and Proceedings	Mao XW, Stanbouly S, Jones T, Nelson G. "Ocular response in the retina and optic nerve head after single and fractionated low-dose high-energy protons." 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. , Feb-2022
Articles in Peer-reviewed Journals	Mao XW, Stanbouly S, Chieu B, Sridharan V, Allen AR, Boerma M. "Low dose space radiation-induced effects on the mouse retina and blood-retinal barrier integrity." Acta Astronaut. 2022 Jul 30;199:412-19. https://doi.org/10.1016/j.actaastro.2022.07.029 , Jul-2022